Day 22 Task: Security Log Fundamentals

Security logs are records of activities and events generated by operating systems, applications, servers, and network devices. They provide valuable information about system activity, user actions, authentication attempts, errors, and potential security incidents.

Security log analysis is an important part of cybersecurity because it helps identify suspicious behavior, investigate incidents, detect unauthorized access, and understand what happened during a security event.

What Is a Security Log?

A security log is a record of events relevant to the security and operation of a computer system or network. These records may include successful logins, failed authentication attempts, permission changes, account modifications, and access to protected resources.

Security professionals examine logs to identify unusual patterns, investigate suspicious activity, and determine whether a system may have been compromised.

Examples of security logs include authentication logs, system logs, firewall logs, web server logs, and application logs.

Common Types of Security Logs

Authentication Logs

Authentication logs record attempts to access a system, including successful logins, failed logins, and authentication-related events. Repeated failed login attempts may indicate a brute-force attack or an incorrectly configured application.

System Logs

System logs record operating system activities, service events, warnings, errors, and other system-related information. They help administrators investigate system failures and unusual behavior.

Firewall Logs

Firewall logs record network traffic events that the firewall allows or blocks. They may contain source and destination IP addresses, port numbers, protocols, timestamps, and the action taken by the firewall.

Application Logs

Application logs record events generated by software applications. They may include user actions, application errors, authentication activity, and access to application resources.

Web Server Logs

Web server logs record requests made to web servers. They commonly include client IP addresses, requested URLs, timestamps, HTTP status codes, and user-agent information.

Important Security Log Fields

Timestamp

The timestamp indicates when an event occurred. It helps analysts establish the sequence of events and correlate activity across different systems. Time zones and clock synchronization should be considered during investigations.

Username

The username identifies the account associated with an event, when available. It helps investigators determine which account was involved in a login attempt or other activity.

Source IP Address

The source IP address identifies the network address from which a connection or request originated, as observed by the logging system. It can help trace suspicious traffic, although proxies, VPNs, NAT, and spoofing can affect attribution.

Event Type

The event type describes the activity recorded, such as a successful login, failed login, permission change, or access denial.

Status or Result

This field indicates whether an operation succeeded or failed. For example, an authentication event may be recorded as SUCCESS or FAILURE.

Event ID

Some logging systems assign an event ID or unique identifier to a particular event type. This helps analysts identify and filter relevant records.

Example of a Security Log

The following is a synthetic authentication log created for learning purposes.

2026-10-22 09:15:20 user=alice src_ip=192.168.1.10 event=LOGIN status=SUCCESS

2026-10-22 09:17:04 user=bob src_ip=203.0.113.25 event=LOGIN status=FAILURE

2026-10-22 09:17:15 user=bob src_ip=203.0.113.25 event=LOGIN status=FAILURE

2026-10-22 09:17:29 user=bob src_ip=203.0.113.25 event=LOGIN status=FAILURE

2026-10-22 09:20:11 user=admin src_ip=192.168.1.20 event=LOGIN status=SUCCESS

In this example, Alice successfully logs in from a private network address. Bob has three failed login attempts from the same source IP address within a short period. This pattern may warrant investigation, although it could also result from a forgotten password or another legitimate issue.

The administrator account subsequently logs in successfully. An analyst would examine additional logs and context to determine whether the activity is expected.

How to Analyze Security Logs

Security log analysis involves examining records to identify meaningful events and suspicious patterns.

- Collect logs: Gather relevant records from systems, applications, and network devices.
- Identify important fields: Locate timestamps, usernames, source IP addresses, event types, and status values.
- Filter events: Search for failed logins, unusual access attempts, permission changes, or other events relevant to the investigation.
- Identify patterns: Look for repeated failures, unexpected login times, unfamiliar source addresses, or unusual account activity.
- Correlate events: Compare records from multiple sources to understand the sequence of activities.
- Document findings: Record suspicious events, supporting evidence, and recommended actions.

A single unusual event does not necessarily indicate an attack. Analysts should evaluate patterns and supporting evidence before reaching conclusions.

Basic Linux Commands for Log Analysis

Linux provides commands that help administrators inspect log files.

View a log file

cat auth.log

Displays the contents of a file named auth.log in the current directory.

Display the latest log entries

tail -n 20 auth.log

Displays the last 20 lines of the file.

Search for failed login events

grep "FAILED" auth.log

Searches for lines containing FAILED, assuming the log uses that exact text.

Count failed login entries

grep -c "FAILED" auth.log

Counts the lines containing FAILED.

Monitor new log entries

tail -f auth.log

Displays new lines as they are appended to the file.

Actual Linux authentication log locations and formats vary by distribution. Some systems use /var/log/auth.log, others use /var/log/secure, and many use the system journal, which can be inspected with journalctl. Access may require administrator privileges.

Why Security Logs Are Important

Security logs support threat detection, incident response, troubleshooting, and compliance activities. They help analysts identify suspicious login attempts, investigate unauthorized changes, and reconstruct events that occurred before or during a security incident.

Centralized logging and security monitoring tools can combine records from multiple systems to reveal patterns that might not be visible in a single log file.

Logs should be protected against unauthorized modification and retained according to the organization's security and privacy requirements.

Best Practices for Security Log Management

- Enable logging for important security events.
- Synchronize system clocks to support accurate event correlation.
- Restrict access to log files.
- Protect logs against unauthorized alteration or deletion.
- Monitor repeated failed logins and unusual account activity.
- Centralize logs when appropriate.
- Avoid recording passwords, authentication secrets, or unnecessary sensitive information.
- Review logs regularly and investigate suspicious patterns.
- Establish suitable retention and backup policies.

Security logs are essential for understanding system activity and identifying potential cybersecurity threats. By examining timestamps, usernames, source IP addresses, event types, and status values, analysts can recognize suspicious patterns and investigate incidents. Effective logging, secure storage, regular reviews, and proper event correlation help organizations improve their security monitoring capabilities.